Does Structured Debate Reduce Moral Bias in Large Language Models?

Aim

We compare standard single-response against structured debate at the model level. The two core outcomes are yes-no bias and omission bias, where lower values indicate smaller framing gaps.

Key Takeaways

1. DeepSeek and Alibaba Qwen3.6 show the most stable support: debate reduced both bias types in completed Exp2 and Exp3 runs.
2. Qwen local, Gemma 4 31B, and Gemini 2.5 Flash show mixed outcomes, indicating that debate gains are strongly model-dependent.
3. Zhipu GLM-5.1 improved on average in Exp3, but the lower debate validity rate weakens the robustness of that result.

Statistical Note

The framing x prompt ANOVA interaction is significant for every included run.
The strongest evidence appears in DeepSeek and Alibaba Qwen3.6, both with $p < .001$.
Zhipu GLM-5.1 / Exp3 is also significant, but the debate valid-response rate is only 0.914.

Methods Note

We used the original OSF Exp2/Exp3 stimuli.
Each dilemma x framing x condition cell used 56 repeated samples.
Only completed and analyzed model-level runs are included.

Model	Dataset	Valid Std / Deb	Yes-no Std -> Deb	Omission Std -> Deb	Takeaway
DeepSeek	Exp2	1.000 / 1.000	0.423 -> 0.196	0.530 -> 0.256	Strong support
DeepSeek	Exp3	1.000 / 1.000	0.833 -> 0.375	0.667 -> 0.467	Strong support
Alibaba Qwen3.6	Exp2	1.000 / 1.000	0.155 -> 0.045	0.321 -> 0.286	Strong support
Alibaba Qwen3.6	Exp3	1.000 / 1.000	0.557 -> 0.244	0.500 -> 0.220	Strong support
Zhipu GLM-5.1	Exp3	1.000 / 0.914	0.726 -> 0.504	0.833 -> 0.417	Promising but less robust
Qwen local	Exp2	1.000 / 0.999	0.360 -> 0.327	0.027 -> 0.223	Partial / mixed
Gemma 4 31B	Exp2	1.000 / 1.000	0.167 -> 0.012	0.167 -> 0.214	Partial / mixed
Gemini 2.5 Flash	Exp2	1.000 / 0.999	0.036 -> 0.178	0.202 -> 0.107	Partial / mixed